

SUPPLEMENTARY MATERIAL: PREDICTIVE MEMORY LOCALIZATION

Anonymous authors

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S1 FROZEN BENCHMARK CONSTRUCTION

The frozen benchmark is constructed from nine public multiple-choice and reasoning sources (Table S1). Records retain their source dataset, domain, release year, and freshness group. The final collection contains 1,950 records from 2024 sources and 1,050 records from pre-2024 sources. Fourteen domains range from situated commonsense and elementary science to recent corrected facts, academic questions, mathematics, and reasoning.

The source benchmarks are MMLU-Pro and MMLU-Redux 2.0 (Wang et al., 2024; Gema et al., 2025), AI2 ARC, OpenBookQA, and SciQ (Clark et al., 2018; Mihaylov et al., 2018; Welbl et al., 2017), LiveBench (White et al., 2025), HellaSwag (Zellers et al., 2019), and QASC (Khot et al., 2020).

Table S1: Source composition of the frozen 3,000-record benchmark.

Source dataset	Records
MMLU-Pro	1,029
MMLU-Redux 2.0	571
AI2 ARC	350
OpenBookQA	200
SciQ	200
LiveBench reasoning	200
HellaSwag	150
QASC	150
LiveBench math	150
Total	3,000

Each record contains a direction-fitting set and an evaluation assignment. The direction-fitting statements never contain the target, neighbor, or capability evaluation fields. Evaluation uses three target probes, three semantic-neighbor probes, and four capability probes per record. Capability assignments are round-robin balanced over 2,834 unique probes; every capability probe is used four or five times. This prevents a small set of easy generic questions from dominating the capability-damage estimate.

S1.1 INDEPENDENT EXPERT SEMANTIC AUDIT

Multiple human experts independently reviewed a dataset-stratified sample of 108 complete records, 12 from each source. They checked the desired and contrast answers, direction-fitting statements, three target probes, three semantic neighbors, and four capability probes. Sixty-five records pass without issue, 32 have a minor issue, and 11 fail at least one semantic criterion. This yields an 89.8% acceptable rate on the audited sample. We additionally remove the known failed records and then all known non-pass records from the full CPU analyses. Table S2 shows that principal predictor AUROC changes by at most 0.004. At block 7, all learned Target and Clean gains over random retain positive paired 95% intervals under both exclusions; for RFM/AGOP, the most conservative exclusion gives +0.0369 Target and +0.0352 Clean. Independently authored validation remains future work.

Table S2: Sensitivity to the independent expert semantic audit. The stratified audit covers 108 records (12 per source): 65 pass, 32 minor issue, and 11 fail. Cells are complete-predictor AU-ROC/AP after retaining all records, excluding the 11 fails, or excluding all 43 non-pass records.

Records retained	Target	Clean	N-dmg.	C-dmg.
All 3,000	.815/.357	.810/.330	.858/.396	.856/.382
Exclude fails (2,989)	.818/.369	.813/.341	.858/.392	.855/.384
Exclude non-pass (2,957)	.817/.368	.813/.344	.857/.398	.853/.391

S2 CONFIRMATORY EXPERIMENTAL DETAILS

We evaluate Qwen3-1.7B-Base Yang et al. (2025) at 0-indexed layers 7 and 11, selected from a dense pilot over layers 7, 11, 15, 19, 23, and 27. The main grid is $\{-0.5, -0.25, -0.1, 0, 0.1, 0.25, 0.5\}$. Five distinct direction families plus a matched-random audit entry, which is numerically identical to the random control, and two layers produce 30,000 distinct record-method-layer paths and 36,000 logged rows. Seven coefficients and ten probes per record produce 210,000 distinct and 252,000 logged path-strength rows, plus 2.52 million logged probe-level rows. Equivalently, the audit artifacts contain 360,000 seven-point probe trajectories, of which 300,000 are distinct after removing the duplicate entry. GPU inference uses one NVIDIA GeForce RTX 5090 with bfloat16, batch size two, and seed 113.

Direction construction and inference. Each direction is fitted independently for one record and layer using 6–13 positive and 6–13 contrast prompts. Prompts are tokenized without added special tokens and left padded; the direction-fitting activation is the final non-padding prompt token at the selected layer. Mean difference, ridge linear, and logistic use the same activation matrix and labels. Ridge regularization is 10^{-3} , logistic regression runs for at most 1,000 iterations, and RFM uses three iterations with a Laplace kernel of bandwidth 10 and regularization 10^{-3} . During evaluation, the unit direction is added to the selected block output at every token position. Correct and contrast continuations are teacher-forced separately, and path labels use their mean per-token log-probability margin to reduce continuation-length effects.

Random-direction 95th percentiles define separate response thresholds: $\tau_T = 0.1753$, $\tau_N = 0.1567$, and $\tau_C = 0.1248$. The confirmatory labels use only $|\alpha| \in \{0.25, 0.5\}$. Weak-response features use only $|\alpha| = 0.1$, so the observed probe strength is disjoint from all label strengths.

Feature group B contains unperturbed target, neighbor, and capability margins. Group M contains method, layer, dataset, domain, freshness, and release year. Group L contains separation, saliency, threshold-accuracy, and direction-agreement statistics, while G contains RFM/AGOP spectrum and alignment statistics. Group R contains signed target, neighbor, and capability responses at $\alpha = \pm 0.1$. Prediction uses class-balanced logistic regression and random forests with five grouped folds by record, dataset, or domain. We report AUROC, average precision, balanced accuracy, F1, and Brier score in the released result artifacts; the tables below retain AUROC and AP, the two threshold-independent ranking metrics.

Computational environment. Experiments and paper-facing analyses run on Ubuntu 22.04.5 LTS with an Intel Xeon Gold 6530 CPU (14 allocated cores), 117 GiB RAM, and one NVIDIA GeForce RTX 5090 GPU (driver 580.65.06; CUDA toolkit 13.0). The software environment uses Python 3.10.20, PyTorch 2.12.0+cu132, Transformers 5.8.1, scikit-learn 1.7.2, pandas 2.3.3, NumPy 2.2.5, and SciPy 1.15.3.

S3 DIMENSION-CORRECTED CROSS-MODEL PROTOCOL

The confirmation uses one seed-113 stratified subset of 500 frozen records for Qwen3-1.7B-Base, Qwen3.5-2B-Base Qwen Team (2026), and Ministral-3-3B-Base Liu et al. (2026). A separate seed-127 calibration set contains 100 records and has zero overlap with the formal subset. At each aligned layer, 1,941 direction-construction prompt states are used to estimate the median residual RMS. The calibration set is used only to fix model-layer scales, not to select records, methods, or outcomes.

Table S3: Record-paired percentage-point differences from random in the frozen 3,000-record study, with 95% intervals from 10,000 record bootstrap resamples. Positive target/clean differences are favorable; positive damage differences are unfavorable.

Block 7				
Direction	Target	Clean	N-dmg.	C-dmg.
Mean difference	+3.3 [2.0, 4.6]	+3.2 [1.9, 4.4]	−0.3 [−1.3, 0.7]	+0.0 [−1.1, 1.1]
Linear	+2.1 [0.8, 3.4]	+2.0 [0.7, 3.2]	+0.6 [−0.4, 1.7]	+0.2 [−0.9, 1.3]
Logistic	+2.8 [1.6, 4.1]	+2.8 [1.5, 4.0]	+0.9 [−0.2, 1.9]	+0.8 [−0.3, 1.9]
RFM/AGOP top-1	+3.6 [2.3, 4.9]	+3.4 [2.1, 4.8]	+0.3 [−0.7, 1.3]	+0.8 [−0.4, 1.9]
Block 11				
Mean difference	+2.5 [1.2, 3.8]	+2.3 [1.0, 3.5]	+0.8 [−0.2, 1.8]	+0.0 [−1.0, 1.0]
Linear	+1.4 [0.2, 2.7]	+1.4 [0.2, 2.6]	−0.2 [−1.2, 0.7]	+0.0 [−1.0, 1.0]
Logistic	+2.4 [1.2, 3.7]	+2.3 [1.1, 3.6]	+0.4 [−0.6, 1.4]	+0.0 [−1.0, 1.0]
RFM/AGOP top-1	+1.4 [0.2, 2.7]	+1.5 [0.3, 2.7]	+0.5 [−0.5, 1.5]	+0.1 [−0.9, 1.2]

Directions have unit ℓ_2 norm. We therefore match the intervention relative to the residual-state ℓ_2 norm when hidden widths differ: $\|\alpha v\|_2 / \|h\|_2$ with

$$s_{m,\ell} = \frac{\text{RMS}(h_{m,\ell})}{\text{RMS}(h_{\text{ref},\ell})} \sqrt{\frac{d_m}{d_{\text{ref}}}}, \quad \alpha_{m,\ell} = s_{m,\ell} \alpha_{\text{ref}}. \quad (\text{S1})$$

Qwen3-1.7B and Qwen3.5 have hidden width 2,048; Ministral has width 3,072 and therefore includes a $\sqrt{3,072/2,048}$ correction. The resulting layerwise scales are 1.000/1.000 for Qwen3-1.7B layers 7/11, 0.088/0.041 for Qwen3.5 layers 6/9, and 0.091/0.052 for Ministral layers 6/10. The manifest stores the RMS ratio, hidden-width ratio, scale, record hash, and source-summary hashes.

All models retain the same target, neighbor, and capability probes; reference coefficient grid; and confirmatory outcome definitions. At both aligned layers we evaluate random, mean difference, logistic, and RFM/AGOP, for 24 model–layer–method configurations, 12,000 record-level paths, and 84,000 path–strength evaluations. Linear remains in the complete 3,000-record primary study but is omitted here because logistic represents the same supervised discriminative direction family. Each model recalibrates target, neighbor, and capability thresholds from its own random directions on the frozen 500-record cohort. Qwen3-1.7B has scale one, so its 500-record outcomes are exact subsets of the completed primary run; only the cohort-level random thresholds are recomputed.

The corresponding path-incidence outcomes are reported in the main paper. This section records the calibration and normalization choices needed to reproduce that comparison without duplicating the main result table.

S4 UNCERTAINTY AND WORKED EXAMPLES

For Table S3, each learned row is paired with the random row for the same record and layer. We resample the 3,000 record identifiers with replacement 10,000 times and recompute the mean paired difference; all method-specific measurements from a sampled record remain together. The table reports percentile intervals from the deterministic bootstrap implemented in the paper asset script.

S4.1 WORKED FROZEN RECORD AND PATH LABELS

The main paper traces one unchanged record from direction-fitting statements to target, neighbor, capability, and path labels. Table S4 adds clean-only, collateral-without-clean, no-effect, and mixed-sign examples. The mixed example also illustrates why Clean and damage-any are not complements: one sign can contain a clean operating coefficient while collateral movement occurs elsewhere on the full path.

Table S4: Additional frozen-record path examples. Onsets list the first suppression/enhancement/damage threshold crossings; a dash denotes no crossing. The examples illustrate label construction rather than quantitative evidence.

Type	Record (source)	Path	Onsets S/E/D	Clean
Mixed	13c polarization relaxation time (MMLU-Redux)	Mean diff., 11	-.25/-+.50	1
Clean only	ferret beverage soy milk (reasoning)	RFM, 11	-/+ .50/-	1
Collateral	coal mines energy (Open-BookQA)	RFM, 7	-/+ .25/+ .25	0
No effect	acid spill water 01 (AI2 ARC)	Logistic, 7	-/-/-	0

Table S5: Macro AUROC across the eight outcomes visualized in the main paper. ΔL and ΔR are cumulative gains; Final is $B + M + L + R$. Dataset and Domain use the final predictor. The Qwen3-1.7B subset comes from the primary cohort.

Cohort	$B+M$	ΔL	ΔR	Final	Dataset	Domain
Qwen3-1.7B primary	0.686	+0.2	+16.1	0.849	0.843	0.840
Qwen3-1.7B subset	0.612	-0.1	+21.7	0.828	0.829	0.828
Qwen3.5-2B	0.621	+1.3	+17.1	0.805	0.806	0.800
Ministral-3B	0.615	+2.2	+16.4	0.801	0.796	0.800

S5 ADDITIONAL PREDICTION ANALYSIS

Across all cohorts, split types, targets, and both prediction models, adding the disjoint weak probe R provides the main predictive gain; we report split-specific random-forest results in Table S8. The frozen primary all-method prediction artifacts use 36,000 logged rows, including the matched-random audit entry, alongside 30,000 distinct paths. Grouping by record keeps the audit entry within its fold and preserves the logged-cohort estimate. Geometry G is outcome-specific within the RFM cohort: it improves later enhancement and target-any prediction but reduces capability-damage AUROC.

The main paper reports the cross-model summary. Here we provide the principal CPU controls, full layer-selection exclusion, weak-response baseline, grouped-transfer, and RFM-specific geometry results.

S5.1 MEASURED-GRID PATH TOPOLOGY

We recompute descriptive topology on all nonzero measured strengths $|\alpha| \in \{0.1, 0.25, 0.5\}$. Across learned directions, a target crossing exists on 7.34% of paths, a clean coefficient on 6.94%, and a nonmonotonic target-or-damage indicator on 10.90%; the corresponding random rates are 5.98%, 5.55%, and 10.65%. Strict target-first and damage-first patterns are rare (0.37% and 0.46% for learned directions), so they serve as descriptive diagnostics alongside the primary prediction targets. Among learned paths with any clean coefficient, 76.3% contain exactly one measured clean coefficient. These results support evaluating multiple strengths on the measured grid.

S5.2 LAYER-SELECTION EXCLUSION AND WEAK-RESPONSE CONTROLS

The 500 records used to select the two primary blocks are a subset of the 3,000-record cohort. Removing them leaves 2,500 records and 30,000 method-block paths. At block 11, all four learned direction families retain positive paired Target-any and Clean-any differences from random (Table S6). The complete record-held-out random forest also remains within 0.01 AUROC of its full-cohort estimate on each principal outcome (Table S7). The exclusion analysis therefore preserves both the outcome and prediction estimates after removing the layer-selection subset.

The training-free scalar control uses only the signed $|\alpha| = 0.1$ response matched to each later outcome. The R -only random forest instead uses the multivariate low-dose response profile. For

Table S6: Sensitivity after excluding the 500-record layer-selection subset. Entries are learned-minus-random path-incidence differences at block 11 with record-paired 95% bootstrap intervals on the remaining 2,500 records.

Direction	Target-any	Clean-any
Mean difference	.034 [.021,.048]	.033 [.019,.047]
RFM/AGOP	.034 [.020,.048]	.032 [.019,.046]
Logistic	.026 [.012,.040]	.024 [.011,.038]
Linear	.016 [.003,.029]	.016 [.003,.028]

Table S7: Record-held-out CPU prediction controls on Qwen3-1.7B-Base. In panel (a), each cell is AUROC/AP on all 3,000 records. Scalar is the training-free signed weak-response score; *R*-only and Full are random forests, where Full uses $B + M + L + R$. Panel (b) reports prevalence, AUROC, and AP with 95% intervals after excluding the 500-record layer-selection subset.

(a) Scalar and multivariate weak-response controls				
Outcome	Prev.	Scalar	<i>R</i> -only	Full
Supp.	.058	.793/.296	.843/.338	.865/.327
Enh.	.056	.771/.282	.833/.320	.852/.312
Target-any	.111	.708/.324	.793/.367	.815/.357
Neighbor dmg.	.079	.840/.408	.842/.402	.858/.396
Capability dmg.	.078	.849/.373	.842/.373	.856/.382
Clean supp.	.054	.729/.227	.841/.308	.863/.299
Clean enh.	.053	.735/.239	.832/.305	.851/.292
Clean-any	.105	.651/.266	.789/.342	.810/.330

(b) Complete random forest after subset exclusion			
Outcome	Prev.	AUROC [95% CI]	AP [95% CI]
Target-any	.101	.824 [.810,.838]	.355 [.298,.412]
Clean-any	.095	.819 [.806,.831]	.328 [.278,.377]
Neighbor dmg.	.075	.858 [.848,.868]	.392 [.369,.414]
Capability dmg.	.078	.852 [.850,.856]	.373 [.338,.400]

target and clean outcomes, this profile substantially improves over the scalar score; adding base, method/block, and static-localization features provides a smaller final AUROC gain. We report prevalence and AP alongside AUROC because the positive path labels are sparse. AP varies across outcomes, while the structured weak response consistently adds ranking information to the prediction task.

S5.3 GROUPED TRANSFER AND GEOMETRY

S6 MULTI-FIDELITY STRENGTH SELECTION

The dense training study contains 500 records, three methods, six layers, and 27 coefficients, yielding 243,000 raw record-strength rows. A disjoint 100-record validation study contributes 24,300 raw dense rows. Excluding the zero coefficient leaves 234,000 dense training candidates and 23,400 validation candidates. Multi-fidelity training adds 2,400 non-overlapping records with sparse trajectories, for 2,900 training records and 320,400 nonzero candidate rows; validation overlap is zero.

We evaluate selection by held-out policy replay. The global fixed coefficient is chosen only from mean utility on the 500-record dense training set, which selects -1 for suppression and $+1$ for enhancement. For every unseen validation path, a learned policy observes metadata, the candidate coefficient, and the two $|\alpha| = 0.1$ weak responses, then either chooses one coefficient or abstains. The measured dense-grid outcome at the selected coefficient is revealed only after this decision, keeping the validation response curve outside selector and fixed-baseline tuning.

Table S8: Macro performance over the eight confirmatory later-strength outcomes. Values are random-forest AUROC/AP in the all-method cohort.

Split	Features	AUROC	AP
Record	$B + M$	.686	.144
Record	$B + M + L$	.688	.145
Record	$B + M + R$	.848	.333
Dataset	$B + M$	.645	.129
Dataset	$B + M + L$	.644	.126
Dataset	$B + M + R$	.841	.305
Domain	$B + M$	.642	.125
Domain	$B + M + L$	.641	.123
Domain	$B + M + R$	.838	.303

Table S9: RFM-cohort geometry ablation, averaged over record-, dataset-, and domain-grouped splits. G contains AGOP spectrum and alignment features. Deltas compare $B + M + L + G$ with the $B + M + L$ base.

Outcome	Base	+ G	Δ AUC	Δ AP
Suppression	.651	.646	-.005	-.007
Enhancement	.645	.680	+.035	+.005
Target-any	.666	.688	+.022	+.014
Neighbor damage	.632	.635	+.003	+.008
Capability damage	.622	.589	-.033	-.021
Clean suppression	.652	.649	-.002	-.003
Clean enhancement	.644	.680	+.037	-.002
Clean-any	.666	.690	+.025	+.018

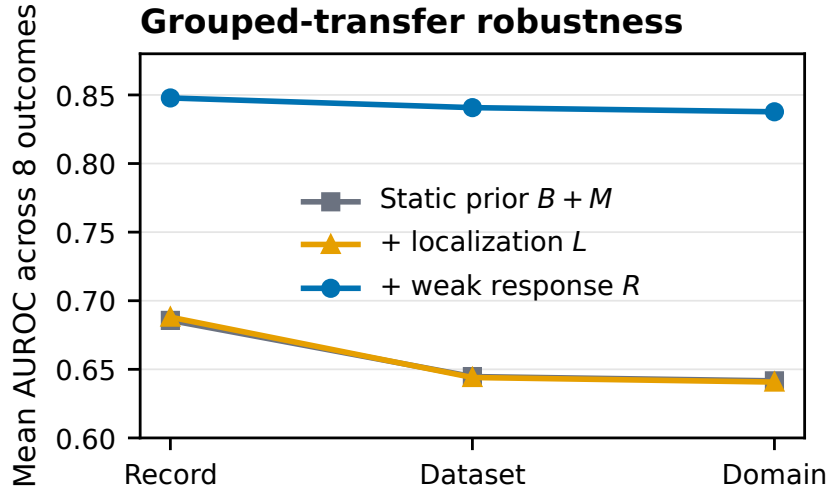


Figure S1: Grouped-transfer robustness. Static localization leaves the baseline nearly unchanged, whereas the disjoint weak response produces a large AUROC gain under record-, dataset-, and domain-held-out evaluation.

Without a weak response, the multi-fidelity $B + M + A$ gradient-boosted decision tree (GBDT) reaches clean-alpha AUROC 0.610 and AP 0.087. Adding R raises the $M + A + R$ model to AUROC 0.849 and AP 0.323; adding B or L after R adds no consistent ranking gain. For the same $M + A + R$ model, multi-fidelity training increases held-out clean rate over dense-only training by 1.0 point for suppression and 0.8 points for enhancement, while utility changes by less than 0.001.

Table S10: Held-out dense-grid strength selection on 100 records. A denotes candidate-strength features; MF denotes multi-fidelity training.

Suppression				
Selector	Clean	Utility	N-dmg.	Abstain
No intervention	.000	.000	.000	1.000
Fixed $ \alpha = 1$	.098	-.041	.060	.000
Dense $M + A + R$	.106	.013	.016	.473
MF $B + M + A$	.060	-.018	.037	.447
MF $M + A + R$	.116	.013	.018	.454
MF $B + M + A + R$	.110	.012	.018	.483
MF $B + M + L + A + R$	.098	.011	.019	.471
Dense oracle	.232	.084	–	–

Enhancement				
Selector	Clean	Utility	N-dmg.	Abstain
No intervention	.000	.000	.000	1.000
Fixed $ \alpha = 1$	.119	-.026	.052	.000
Dense $M + A + R$	.109	.008	.024	.407
MF $B + M + A$	.079	-.017	.042	.312
MF $M + A + R$	.117	.008	.022	.391
MF $B + M + A + R$	.112	.010	.024	.422
MF $B + M + L + A + R$	.113	.010	.026	.417
Dense oracle	.260	.089	–	–

Table S11: Expected intervention evaluations per held-out path. Learned policies use two weak probes and execute the selected coefficient only when they do not abstain. Reduction is relative to evaluating all 26 nonzero dense-grid coefficients.

Policy	Suppression	Enhancement
Train-tuned fixed	1.000	1.000
Dense-only $M + A + R$	2.527	2.593
Multi-fidelity $M + A + R$	2.546	2.609
Dense scan	26.000	26.000
Multi-fidelity reduction	90.2%	90.0%

The practical gain comes from risk-aware selection rather than uniformly higher clean rates (Table S10). No intervention has zero utility by construction, while the train-tuned fixed $|\alpha| = 1$ policy has negative utility in both directions. The multi-fidelity $M + A + R$ selector abstains on 45.4% of suppression paths and 39.1% of enhancement paths, reducing neighbor damage from 6.0% to 1.8% and from 5.2% to 2.2%, respectively. Both paired utility gains over the fixed coefficient are significant. Suppression utility is also significantly positive relative to no intervention, whereas the enhancement interval against zero overlaps zero (Table S12). Its expected cost is 2.55 and 2.61 intervention evaluations per path, about 90% fewer than the 26-point nonzero dense scan (Table S11). A substantial oracle gap remains; the selector reduces evaluations while dense evaluation remains useful for final audit.

Selector uncertainty uses records as the sampling unit. We first average the nine method–layer paths within each of the 100 validation records, then resample records 10,000 times. Utility and damage comparisons are paired against the train-tuned fixed policy on the same validation record. Table S13 varies one utility weight at a time and retrains the utility head. All seven configurations retain positive paired gains over the fixed policy and reduce neighbor damage. The declared weights are one stable configuration for this within-study risk objective; deployment requires recalibration.

Table S12: Record-paired uncertainty for the multi-fidelity $M + A + R$ selector on 100 held-out records. Brackets are 95% intervals from 10,000 record bootstrap resamples. Outcome changes are selector-minus-fixed in percentage points.

Metric	Suppression	Enhancement
Utility vs. no intervention	.013 [.003,.025]	.008 [−.001,.019]
Utility vs. fixed	.055 [.044,.066]	.034 [.024,.044]
Target change	0.9 [−1.3,3.0]	−0.8 [−2.6,1.1]
Clean change	1.8 [0.0,3.8]	−0.2 [−1.9,1.8]
Neighbor-dmg. change	−4.2 [−6.3,−2.3]	−3.0 [−4.9,−1.4]
Capability-dmg. change	−1.1 [−2.0,−0.4]	−0.2 [−0.7,0.2]

Table S13: Selector utility sensitivity on the same 100 held-out records. Each row retrain the utility head after changing one declared weight. Entries are paired utility gains for suppression/enhancement.

Configuration	ΔU vs. fixed	ΔU vs. zero
Damage weight 0.5	.030/.015	.021/.020
No clean bonus	.056/.041	.015/.015
No strength penalty	.050/.032	.009/.006
Declared weights	.055/.034	.013/.008
Clean bonus 0.2	.041/.026	.000/.000
Strength penalty 0.02	.056/.037	.015/.011
Damage weight 2.0	.113/.090	.008/.003

S7 FREE-GENERATION ENDPOINT STRESS TESTS

The free-generation studies cover four directions, two aligned layers, and coefficients from -1 to 1 . On Qwen3.5, the unmatched raw-alpha stress test changes roughly half of nonzero-strength target generations and produces several learned wrong-to-right transitions, while target damage balances or exceeds these corrections. On Qwen3-1.7B, decoded text often changes without a corresponding target correction. These results place the primary margin-level findings in context and motivate future endpoint-oriented evaluation.

S8 ASSET ATTRIBUTION

Figure 1(a,b) contains explanatory imagery and schematic curves rather than measured PML activations. The representation imagery adapts Activation Atlas by Carter et al. (CC BY 4.0); the selected-block illustration in panel (a) adapts Deep Learning Visuals by David V. Godoy (CC BY 4.0). Quantitative figures are generated from the frozen experimental results.

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Table S14: Path-conditioned free-generation endpoint stress tests on 100 records per model. G/D counts unique target prompts with correctness gain/damage at any nonzero coefficient. Text and clause changes are measured against $\alpha = 0$. Qwen3.5 uses the unmatched raw-alpha stress protocol.

Qwen3-1.7B					
Direction	Paths	Base	G/D	Text Δ	Clause Δ
Random	24	5.6%	1/0	37.2%	21.5%
Mean diff.	24	2.8%	0/1	31.9%	19.6%
Logistic	28	4.8%	0/0	25.4%	12.9%
RFM/AGOP	24	6.9%	0/1	33.2%	13.2%
Qwen3.5-2B					
Random	24	8.3%	0/0	45.0%	19.4%
Mean diff.	24	11.1%	2/4	50.3%	30.4%
Logistic	28	8.3%	3/2	52.1%	32.6%
RFM/AGOP	24	8.3%	2/4	47.4%	28.5%

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